A Fast-Response MPPT Charger for the Técnico Solar Boat

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***Abstract*—Solar-powered vessels need effective energy conversion under rapidly changing conditions. The solar panels operate in their maximum power point (MPP) thanks to a maximum power point tracker (MPPT) charger. This work presents the design, simulation, and testing of a fast-response MPPT charger for the low-voltage battery of São Gabriel 01, the latest vessel of Técnico Solar Boat (TSB). The developed prototype is a syncronous boost converter operating at 100 kHz, controlled by a STM32 microcontroller. A perturb-and-observe (P&O) algorithm was selected as the best compromise between tracking speed, accuracy, and complexity. Both the algorithm and the converter were simulated in LTspice and Matlab. A hand-assembled prototype, costing about 125 €, reached a measured efficiency of 96.6–96.8% (including all prototype losses) and an estimated power-stage efficiency of 97-98%. Tracking settled within 100 ms in typical irradiance and temperature steps and within 200 ms in the most demanding tests, with about 99% tracking accuracy. The prototype was tested in laboratory and outdor conditions, following the expected CC-CV charging curve. The board also provides CAN telemetry and a configuration interface for integration in the vessel.**

***Index Terms*—MPPT, perturb and observe, synchronous boost converter, solar energy, solar boat.**

I. Introduction

The share of electricity from renewable sources has risen from 20.82% in 1985 to 31.92% in 2024, of which solar accounted for 6.9% worldwide and 14.5% in Portugal [1]. Its application in vehicles is attractive due to it is modularity, low noise, and low maintenance [2]. In 2018, 29% of total CO2 emissions in the EU came from the transport sector, with 4% coming from the maritime sector alone [3].

In 2015 Técnico Solar Boat (TSB) was created with the ambition to design and manufacture vessels ruining on renewable energies. These vessels compete in the Monaco Energy Boar Challenge, where teams from around the world race and demonstrate their innovations. TSB has manufactured seven vessels to date, with the latest being São Gabriel 01 (SG01), a hydrofoil boat that competes in the Sea Lab class. The vessel is propelled by a 130 kW motor on a 300-600 V bus, powered by hydrogen (still being implemented). To power all electronics, coling system, flight control, telemetry, electric steering, etc. an additional battery is used with lower voltage (30-50.4 V), which is charged by the solar panels.

To charge the battery using solar panels a maximum power point tracker charger need to be used, since the maximum power point (MPP) of the solar panels change with irradiance and temperature. These products track the MPP while charging the battery. Figs. 1 and 2 show that shift. At a cell temperature of 25 °C, a higher irradiance raises the panel current and, with it, the available power. At 1000 $\mathrm{W/m^2}$, a higher cell temperature lowers the panel voltage and therefore the power [4]. A MPPT charger that uses a fixed voltage misses both movements of the MPP.

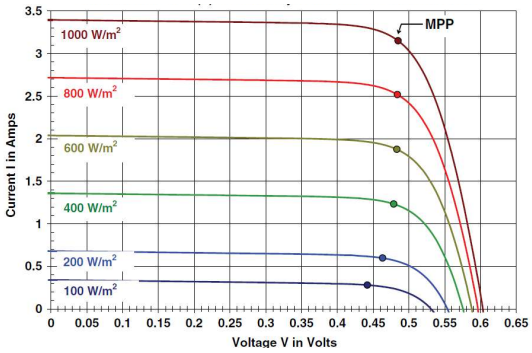


Fig. 1. I–V curves at a cell temperature of 25 °C and variable irradiance [4].

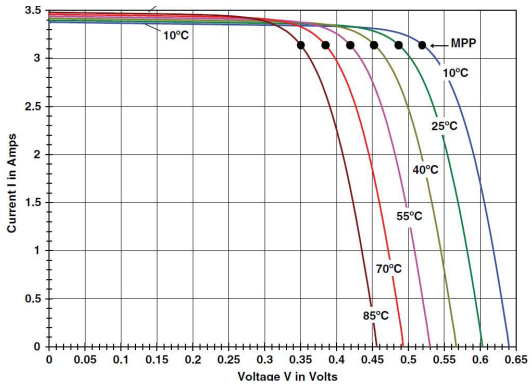


Fig. 2. I–V curves at 1000 $\mathrm{W/m^2}$ and variable cell temperature [4].

Earlier TSB boats used commercial MPPT chargers. An in-house attempt in 2020 worked, but it was never installed because it was not robust, and it no longer matches the team's standards. Units that fit the voltage range are expensive, provide no telemetry, or track slowly. This work therefore builds a fast tracker, a validated power stage, CAN telemetry, and a way to change the configuration.

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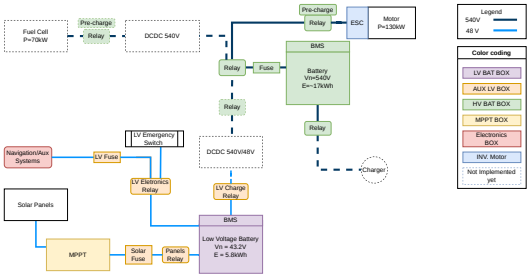


Fig. 3. Simplified electrical architecture of SG01. Solar conversion serves the low-voltage auxiliary battery.

TABLE I
STRING OPERATING POINTS AT THE MPP (MAXEON GEN 3).

String	V_{mpp} (V)	I_{mpp} (A)	P_{mpp} (W)	V_{oc} (V)	I_{sc} (A)
40 cells	24.68	6.05	149.31	29.0	6.374
35 cells	21.88	6.05	132.37	25.4	6.374
20 cells	12.34	6.05	74.66	14.5	6.374

II. SYSTEM

A. Electrical architecture

The electrical system is composed by an low voltage and an high voltage system. The high-voltage system is responsible for propelling the vessel. The low voltage system is responsible for the rest of the electrical components of the vessel, such as the electric string, throttle, telemetry, flight control, etc. A low voltage battery was built by the team using lithium-ion cells in a 12s28p configuration. To increase the autonomy of the low voltage battery, 12 solar panels were built by the team and are attached to the deck, connected to four MPPT chargers, totaling a maximum power of 526.4 W. However, there is still room to grow. Unused deck area could raise that to about 977 W.

Based on the current installation and on previous vessels made by TSB, it was defined that the prototype must be able to use cell strings from 20 to 40 series cells and is optimized for 35 cells (Table I).

The onboard low-voltage pack is a 12s28p assembly of Samsung INR21700-50G cells (43.2 V nominal, 50.4 V maximum, 30 V minimum, 135.8 Ah). It was carried over from another boat and is heavier than SG01 needs. Autonomy estimates therefore also use the smaller 12s7p pack from another vessel TSB vessel, São Rafael 03 (35 Ah, 1512 Wh).

At STC the present array roughly doubles the endurance of the 1512 Wh pack under a 1000 W auxiliary load, from 1 h 30 min with no sun to 3 h 12 min with sun. Higher cell temperature and lower irradiance reduce that gain.

B. How fast the tracker must be

To increase the energy extracted by the MPPT charger, the tracking algorithm must be capable of responding to changes in the operating environment. Many of these changes are unpredictable, such as a reduction in temperature caused by seawater covering the solar panels. However, in a solar vessel, one periodic source of variation can be used as a basis for defining the required tracking speed: the vessel's roll and pitch motion.

The frequency of these oscillations is determined by the wave excitation and the natural frequencies of the hull. An approximation performed for SG01 resulted in natural periods of 0.87 s in pitch and 1.29 s in roll. These values indicate that the vessel response is expected to be amplified when the excitation frequency approaches the corresponding natural frequencies. Furthermore, studies indicate that a significant portion of wave energy is concentrated at periods between approximately 1 and 30 s [5]. Therefore, the vessel is expected to experience oscillations with frequencies within a similar range to those of the incident waves.

To obtain a more representative estimate for the operating conditions of the vessel, several datasets were collected from SG01. The resulting data showed spectral peaks at approximately 0.05 Hz and 0.5 Hz.

Based on these observations, the tracking speed requirement was defined such that the operating point should return to the MPP within 1 s, with a target tracking time of 0.5 s under typical conditions. This requirement ensures that the MPPT algorithm can respond sufficiently quickly to the periodic variations caused by the vessel's motion. In comparison, a tracking time of one to two seconds, as specified by some integrated charge controllers, may result in significant energy losses during rapidly varying operating conditions.

III. STATE OF THE ART

A. MPPT algorithm

Four methods were compared for this charger: constant voltage, perturb and observe, incremental conductance, and a look-up table. The comparison uses the three quantities that matter on SG01: how quickly the operating point returns to the MPP after a change of irradiance or temperature, how close it stays once it is there, and how much computation each update requires.

Constant voltage: The converter is controlled to hold the panel at one chosen voltage. In the simplest form that voltage is a fixed number. A widely used variant sets it to a fraction of the open-circuit voltage, typically 71-78%, or to a fraction of the short-circuit current [6], [7]. The reference is measured by opening or shorting the panel for one sample, or by reading a dummy cell so that the main string keeps producing power [6]. The controller then regulates the panel to that reference. No power calculation is required. Accuracy depends on the assumption that the MPP stays at a fixed fraction of V_{oc} or I_{sc}. Irradiance and temperature both move this fraction [8], so the panel settles offset from the true peak, and every open-circuit sample is an interval with no delivered power.

Perturb and observe: P&O applies a small voltage step and compares the panel power before and after it [9], [10]. The next step follows the signs of the two changes:

$$\begin{aligned} \Delta P > 0 &\Rightarrow \text{repeat the sign of } \Delta V, \\ \Delta P < 0 &\Rightarrow \text{reverse the sign of } \Delta V. \end{aligned} \tag{1}$$

A step that raises the power moved the operating point toward the peak, so it is repeated. A step that lowers the power moved the point away from the peak, so the direction is reversed. Two samples and a comparison are enough, which is why the method is used in most commercial chargers. At the peak the same rule has no stop condition: the next step leaves the MPP, power falls, and the following step returns, so the panel oscillates around the peak. A smaller step shrinks that oscillation and needs more updates to follow a given change [6], [7]. A rise in irradiance between the two samples can also be read as a successful step, so (1) walks away from the MPP [11].

Incremental conductance: On the power-voltage curve the slope is positive on the left of the MPP, zero at the MPP, and negative on the right. Expanding the derivative gives a test that uses only current and voltage [6], [12]:

$$\frac{dP}{dV} = I + V\frac{dI}{dV}. \tag{2}$$

The operating point is located by comparing the incremental conductance dI/dV with the conductance I/V:

$$\begin{aligned} \frac{dI}{dV} &= -\frac{I}{V} && \text{at the MPP,} \\ \frac{dI}{dV} &> -\frac{I}{V} && \text{left of the MPP,} \\ \frac{dI}{dV} &< -\frac{I}{V} && \text{right of the MPP.} \end{aligned} \tag{3}$$

The voltage is increased on the left of the peak, decreased on the right, and held when the two conductances match. The hold condition and variable step size based on the slope values, remove the steady oscillation of P&O. A sudden change of irradiance is also less likely to be taken as a step in the wrong direction, because the decision uses the local slope rather than the sign of ΔP alone [13]. Each update needs a significant computational power. Also, its complexity makes it harder to tune.

Look-up table: A table stores the duty cycle that places the panel at the MPP for each pair of measured voltage and current [7]. The entries come from a panel model or from a laboratory sweep. At run time the controller selects the nearest entry and applies that duty cycle in one cycle. Extra axes, such as temperature or irradiance, raise the accuracy and also the memory and the maintenance. Aging, a repaired string, or a different number of series cells leaves the stored duty cycles wrong until the table is rebuilt [10].

The devolved prototype uses a classic P&O algorithm. It was chosen because of its high performance and low complexity, making it ideal for rapid prototyping. The incremental conductance approach promises better performance and is therefore the strongest candidate for future implementations.

B. Battery charge profile

The stage keeps the panel at the MPP and charges a lithium-ion battery [14]. The method is constant-current constant-voltage (CC–CV), as in most commercial lithium chargers (Fig. 4) [15], [16]. Below the voltage limit the battery charges

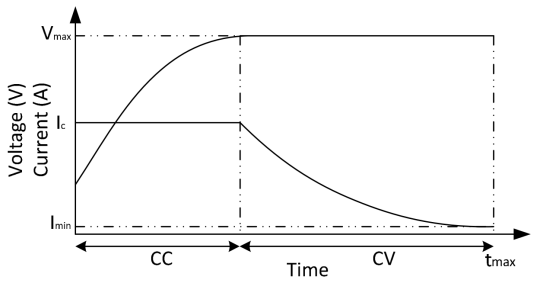


Fig. 4. CC–CV charge curve [14].

at constant current. Since the source is a solar panel with lower current than the maximum battery current, in this stage the MPPT algorithm is employed to provide the maximum current possible to the battery, with a configured maximum. If the output current reaches the maximum battery charging current, a proportional controller is used to maintain this current and the MPPT algorithm steps aside. When the battery voltage reaches the limit, the CC/MPPT algorithm is switched off and enters the constant voltage (CV) stage. In this stage a proportional controller holds the output voltage and lets the current fall. The charge ends once the current drops below a cutoff, which is what stops the pack from overcharging.

Faster lithium profiles shorten the charge at the same current, at the cost of extra switching and complexity [14], [16]. CC–CV is used here for its robustness and simplicity. The laboratory and outdoor charges later in the paper are the measured form of this curve.

C. DC-DC converter

The algorithm sets the panel operating point through a DC–DC converter. Galvanic isolation is not required, so a non-isolated topology is used: a transformer would add size and loss without a function this charger needs.

The three fundamental non-isolated topologies are the buck, boost, and buck-boost converters. Their main difference is their ability to increase or decrease the input voltage. The requirements of this project specify a battery voltage range of 30–50.4 V and a maximum solar panel voltage of 29 V (Table I). Therefore, a boost converter is required to increase the input voltage from the solar panel to the voltage required by the battery.

A classical boost converter stores energy in one inductor while the transistor is on and releases it to the battery through a diode while the transistor is off. Considering ideal devices, the conversion formula between the input and the output is given by:

$$V_{\text{out}} = \frac{V_{\text{in}}}{1-D}, \qquad I_{\text{in}} = \frac{I_{\text{out}}}{1-D}. \tag{4}$$

Considering the diode voltage drop, the transistor drain-source resistance and the inductor resistance, the equation is rewrite as follows:

$$V_{\text{out}} = \frac{V_{\text{in}} - D\,R_{\text{DS(on)}}I_{\text{DS}} - R_L I_L}{1-D} - V_d. \tag{5}$$

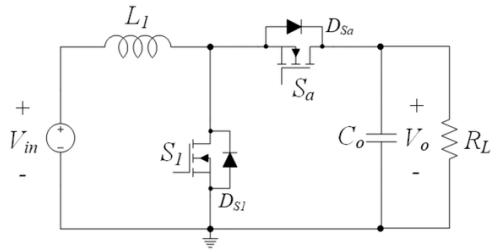


Fig. 5. Synchronous boost stage. The diode of the classical boost is replaced by a controlled MOSFET [17].

The resistive terms reduce the delivered voltage. The diode voltage drop is typically 0.3-0.7 V. Considering the MPP of 35 cells ($V_{mpp} = 21.88\ V$, $I_{mpp} = 6.05\ A$) and the battery at 30 V, ideally the output current is 4.41 A. Considering a diode voltage drop of 0.5 V, the losses caused by the diode are 2.21 W (1.67 % of the converted power). To reduce these losses synchronous boost converter was employed.

The synchronous boost converter (Fig. 5), replaces the diode with a transistor [17]. The term V_d in equation 5 becomes the voltage drop of the transistor. Using the transistor chosen further (section IV), with a $R_{\mathrm{DS(on)}} = 5.2\ \mathrm{m\Omega}$ and the same current, the losses are only 0.1 W (0.07% of the converted power). The inductor equivalent series resistance (ESR) and MOSFET on-resistance remain in (5). These components need to be sized properly in order to reduce the converter losses.

IV. Implementation

A. Power stage

The synchronous boost converter is designed to step up a string of 20 to 40 cells, up to a 12-series lithium-ion battery, 30 to 50.4 V. The switching frequency was defined as 100 kHz. This frequency is reasonable compromise between low ripple voltage and current and high switching losses. Since the battery rail is connected to the vessel electronics, the output voltage ripple goal was defined as $\Delta V_{\mathrm{out}} = 0.1$ V. The input voltage ripple must be low too, so that the readings are not affected.

The components of the converter are chosen to meet the requirements, while also ensuring safety and low losses.

1) Inductor: The inductor size directly influences the inductor ripple current, electromagnetic noise and AC losses. All of these factors are reduced with higher inductance. However, lower inductances can be more efficient by reducing the equivalent series resistance (ESR). Also, the converter can operate in two modes, continuous conduction mode (CCM) and discontinuous conduction mode (DCM). At DCM the inductor current reaches zero which increases the converter losses and noise. Therefore, the inductor must be sized to operate at CCM .

While the low-side transistor is on, the inductor voltage is V_{in} for D/f_{sw} seconds, so:

$$V_L = L \cdot \frac{di_L}{dt} \Leftrightarrow \Delta I_L = \frac{V_{\mathrm{in}} D}{L\, f_{\mathrm{sw}}}. \tag{6}$$

The ripple is written as a fraction of the average inductor current, $r = \Delta I_L / I_{L,\mathrm{avg}}$, with $I_{L,\mathrm{avg}} = I_{\mathrm{out}}/(1-D)$ from (4). Replacing (6) and $V_{\mathrm{in}} = V_{\mathrm{out}}(1-D)$ gives the inductance that meets a chosen ripple:

$$L = \frac{V_{\mathrm{out}}(1-D)^2 D}{I_{\mathrm{out}}\, r\, f_{\mathrm{sw}}}. \tag{7}$$

The duty-cycle factor $(1-D)^2 D$ is largest at $D = 1/3$, where it equals $4/27$. The inductance that still meets the ripple at that worst duty cycle is:

$$L_{\min} = \frac{4}{27}\,\frac{V_{\mathrm{out}}}{I_{\mathrm{out}}\, r\, f_{\mathrm{sw}}}. \tag{8}$$

CCM ends when the current valley touches zero, which is $r = 2$. At the maximum battery voltage, $V_{\mathrm{out}} = 50.4$ V, and at the 10 A (Max(I_{pv})+Margin), (8) then gives

$$L_{\min}(r=2) = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 2 \cdot 100 \times 10^3} = 3.73\ \mu\mathrm{H}. \tag{9}$$

That is the CCM/DCM boundary. An inductor of 3.73 μH runs at the largest ripple while in CCM, with high core loss, device stress, and magnetic interference. The same expression at a ripple factor of 0.4, a peak-to-peak ripple of 40% of the average current, gives a reasonable design reference of:

$$L_{\min}(r=0.4) = 18.7\ \mu\mathrm{H}. \tag{10}$$

After the optimizations in Section V, the inductance of 68 μH (Würth 74437529203680) was chosen. This results in a low ripple factor of $r \approx 0.11$.

The inductor also has to carry the current without overheating or saturating. The rated current must higher than 6.37 A (short-circuit current of the solar panels) with margin, so the specification was set at 10 A. The selected inductor is rated 16 A, with a DC resistance of 10.3 mΩ and a self-resonant frequency of 5.7 MHz. Saturation is specified to 7.2 A for a 10% inductance drop and at 10.6 A for a 30% drop.

The saturation current must be higher than the solar panel short circuit current plus half the ripple current. Considering the open circuit voltage and maximum duty cycle of 100%, the minimum saturation current is calculated as follows:

$$\Delta I_L = \frac{29}{68 \times 10^{-6} \cdot 100 \times 10^3} = 4.26\ \mathrm{A}, \tag{11}$$

$$I_{\mathrm{pk}} = 6.37 + \frac{4.26}{2} = 8.5\ \mathrm{A}, \tag{12}$$

A string cannot be at open-circuit voltage and short-circuit current at the same time, so 8.5 A is an overdimensioned value and not a real operating point. The inductor chosen as slightly lower specification, however if the inductance has already fallen by 30%, the remaining 48 μH is still above the 18.7 μH minimum from (8).

2) Capacitors: During the low-side switch on-time, the output capacitor supplies the battery current. Assuming the input current is the fixed design parameter, the capacitance required to limit the output and input voltage ripples to ΔV_{out} and ΔV_{in}, respectively, are given by:

$$C_{\mathrm{out}} = \frac{I_{\mathrm{in}}(1-D)D}{\Delta V_{\mathrm{out}}\, f_{\mathrm{sw}}}, \qquad C_{\mathrm{in}} = \frac{I_{\mathrm{in}} D}{\Delta V_{\mathrm{in}}\, f_{\mathrm{sw}}}. \tag{13}$$

For a fixed I_{in}, the product $D(1-D)$ reaches its maximum at $D = 0.5$. Substituting $I_{\text{in}} = 10$ A, $\Delta V_{\text{out}} = 0.1$ V, and $f_{\text{sw}} = 100$ kHz into Equation 13 yields a minimum output capacitance of $C_{\text{out}} = 250$ μF. Using the same operating point in the input capacitance expression, with $\Delta V_{\text{in}} = 0.5$ V, gives $C_{\text{in}} = 100$ μF. These values represent the minimum capacitance required to satisfy the specified voltage-ripple limits. The selected capacitors must also meet the voltage-rating and RMS-current requirements.

The input capacitors must withstand the maximum open-circuit voltage of a 40-cell solar panel string, 29 V, plus the input voltage ripple of 0.5 V, resulting in a minimum voltage rating of 29.5 V. Similarly, the output capacitors must withstand the maximum battery voltage of 50.4 V plus the output voltage ripple of 0.1 V, giving a minimum voltage rating of 50.5 V.

The RMS value of the triangular inductor ripple current is given by $(\Delta I_L/2)/\sqrt{3}$. For $\Delta I_L = 4.26$ A, this results in an input current ripple of 1.23 A. As for the output, equation 14 is used calculate the ripple current based on input voltage, inductor ripple and duty cycle (0 maximizes the equation).

$$I_{ripple_{out}} > \sqrt{\frac{I_{\text{in}}^2}{4} + \frac{\Delta I_L^2}{12}(1-D)^2} \tag{14}$$

$$I_{ripple_{out}} = \sqrt{\frac{6.37^2}{4} + \frac{4.26^2}{12}} = 3.42 \text{ A}. \tag{15}$$

A single electrolytic capacitor typically cannot provide 3.4 A of ripple current with a few hundred microfarads, therefore, several capacitors were connected in parallel, which also increase efficiency by reducing ESR.

Two 56 μF were chosen for the input capacitance. Each is rated for 2.4 A, 4.8 A together, and the ESR is 22 mΩ. This capacitor was chosen mainly due to its low ESR. For the output, five 100 μF electrolytic capacitors, each rated for 100 V and 0.8 A. Together, create a 500 μF capacitor bank with 4.0 A rating.

Additionally, several ceramic capacitors were added to filter the high frequency switching noise.

3) Transistors: The transistor is one of the main components of the power converter. Its choice will directly influence the efficiency of the converter. The MOSFETS and GaN transistors are the two main technologies that fit both the frequency and power requirements (150 W and 100 kHz). However, the GaN transistor is the leading edge technologies, standing out due to its fast switching, low drain-source resistance and high power density [18].

The chosen component was the EPC23102, which is a GaN device, that integrates two transistors and its gate drivers, with 5.2 mΩ internal resistance, 100 V, 35 A, in a small package.

B. Sensing and processor

The microcontroller is responsible for controlling the power stage, sensor acquisition, and communication. The chosen microcontroller is an STM32L476 (Cortex-M4, 80 MHz, 1 MB flash, 128 kB RAM), which provides CAN, UART and I2C communication, sixteen 12-bit ADCs and complementary PWM signals. It was chosen because it was already in use in TSB, which limited cost and development risk. A faster clock would increase the duty-cycle precision, and a 16-bit ADC would increase the sensors' precision.

Several safety mechanisms are implemented in the firmware to ensure reliable and safe operation. The most basic protection consists of detecting overvoltage, overcurrent, and overtemperature conditions, which trigger the shutdown of the DC-DC converter for at least 5 s. To prevent shoot-through in the power stage, a dead time of 32 ns is inserted between the complementary PWM signals. Additionally, an independent watchdog timer with a timeout of 1 s resets the microcontroller if it is not refreshed within the specified interval, helping to recover from software faults such as infinite loops or execution failures.

The solar panel and battery voltages and currents are measured and used by the MPPT and charging algorithms. Current sensing is implemented using the INA253A2, an integrated current-sensing device that combines a shunt resistor, a current-sensing amplifier, and a PWM rejection mechanism. The device measures the voltage drop across the integrated shunt resistor and produces an output voltage proportional to the measured current, with a sensitivity of 0.2 V/A. This output is then sampled by the microcontroller's ADC.

Voltage measurements are performed using resistive voltage dividers, which scale the measured voltages to levels compatible with the microcontroller's ADC input range. Using a 3 V voltage reference, the theoretical voltage and current measurement precision is 3.66 mA and 15.38 mV, respectively.

Additionally, the board is capable of measuring the temperature of the transistors, ambient temperature, and solar panel temperature using NTC thermistor.

C. Firmware

A 1 ms tick runs four tasks (Fig. 6): ADC task every 1 ms, the MPPT state machine every 20 ms, CAN task every 100 ms, and serial task every 1 s. The ADC task scales the input reading to the correct units and detects faults. If a fault is detected the Charging state machine is blocked for at least 5 seconds, until the fault is cleared. The MPPT tasks does an average of the last samples and runs the charging state machine. The state machine (Fig. 7) is responsible to decide in which state of the charging curve is in and update the duty cycle accordingly. The ruining period of this task can be changed. Initially 10 ms were used (as in simulations), but it was increased to provide more samples between each step.

The states are divided into charging and non-charging. The non charging contains the delay, disable, No source, No load and Charge finished. The delay lets the output capacitors settle, so that the no load state is no misinterpret. The disable and no source sates just keep the DC-DC turn off. The No load and Charge finished states keep the output voltage at the maximum battery voltage to prevent inrush current when a device is connected. The charge states are Sweep delay, sweep, MPPT algorithm, constant current (CC) and constant voltage (CV).

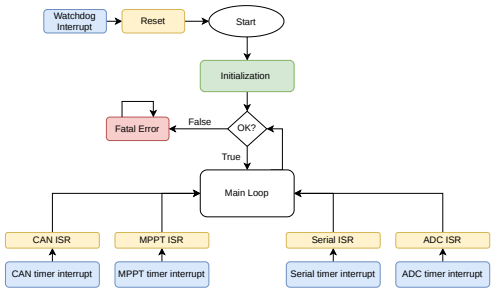


Fig. 6. Firmware flowchart.

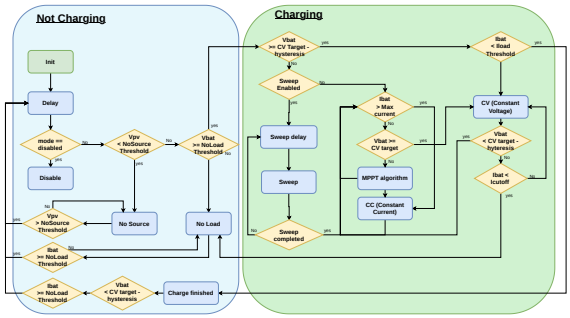


Fig. 7. Charge state machine. Tracking runs only while both the string and the battery are present and inside their limits.

The MPPT algorithm, CC and CV states are responsible for the charging curve, while the sweep and sweep delay are used only one time to find the global MPP. The sweep is optional and is used to ignore local MPPs caused by shading.

The CAN and serial tasks provide two different communication options. The CAN is a widely used protocol in the automotive sector to let all components communicate with in the same bus. It is already in use in TSB vessels and therefore it was implemented. Additionally, the serial protocol was implemented to simplify the configuration process through a Graphical User Interface (GUI).

The GUI was implemented using Qt framework. It provides an accessible user interface to analyze the acquired data and configure the board parameters. The configuration is stored in internal flash by an EEPROM-emulation library and upload on reset.

V. Simulation

A. Converter

The solar panel was modeled in LTspice using a five parameter model (1M5=P) [19], with parameters fitted to V_{oc}, I_{sc}, and the MPP. The battery was a voltage source in series with its resistance (battery internal resistance). Manufacturer models of transistors, capacitors and inductors were used, including the measured rise, fall, and dead times. The converter was run in steady state at a fixed duty cycle, so each run is one operating point rather than a tracking transient.

Every comparison uses the same five points of the CC-CV curve of a 35-cell string at its MPP (Fig. 8). Points 1 to 3 lie on the constant-current interval, where the battery voltage rises from its minimum towards the limit. Here the current is

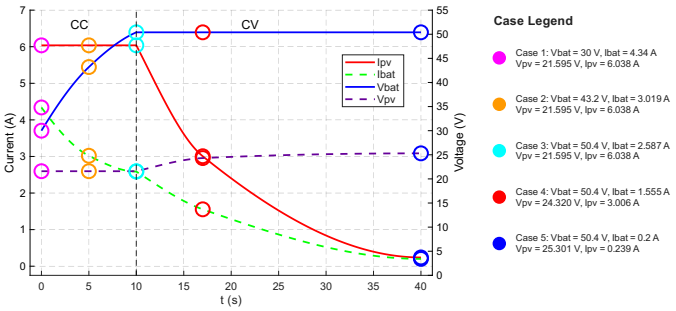


Fig. 8. CC-CV charge of a 35-cell string and the five operating points used in the converter simulations.

TABLE II
LTspice ripple and loss with $L = 68\ \mu$H, two 56 μF input capacitors, and five 100 μF output capacitors.

Point	ΔV_{bat} (V)	ΔV_{pv} (V)	ΔI_L (A)	Loss (%)
1	0.041	0.015	0.98	0.58
2	0.049	0.021	1.71	0.68
3	0.053	0.024	1.96	0.75
4	0.041	0.022	2.01	0.67
5	0.030	0.022	1.96	3.46

set by the string, not by the battery, because the pack accepts more than the string can supply (Section III-B). Point 4 and 5 is in the constant-voltage stage, where the voltage is held and the current drops, and with it the converted power too. Points 1 to 3 therefore test the converter at full power across its conversion ratio, and point 4 and 5 tests it at light load.

The capacitor bank was chosen first, with the inductor held at 68 μH. Four capacitor banks were compared. The lowest loss used two 56 μF inputs and three 220 μF outputs. Those 220 μF capacitors were not available at the moment, therefore, two 56 μF and five 100 μF were chosen, being the second best option. Both options showed losses lower than 1% at the 3 CC points, with a slightly difference of around 0.01%. To minimize costs, using a single 100 μF capacitor at the input and five 100 μF capacitors at the output would be a suitable option. However, this configuration would increase power losses due to the higher equivalent series resistance (ESR) of the capacitor bank.

The inductor was then chosen with those capacitors fixed, sweeping 22 to 100 μH over the same five points. At 22 μH the current ripple reaches 5.6 A, comparable to the solar panel current, and the light-load loss climbs to 6.7%. At 100 μH the ripple drops below 1.3 A, and losses below 3.9%. The 68 μH part sits between the two, with slightly higher ripple current of 1-2 A, the lowest light-load loss of the sweep, and high-power loss within a few hundredths of a point of the smaller inductors. It keeps the inductor ripple low without the copper loss of the 100 μH core.

Table II records the results with the chosen components. Both voltage ripples are inside the 0.1 V and 0.5 V specifications. The fifth point, in the end of the CV stage, loses 3.5%.

LTspice does not split the loss by component. An analytical

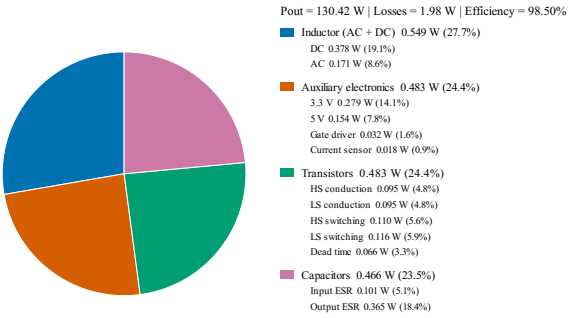


Fig. 9. Analytical loss split at operating point 2. Total loss 1.98 W (98.5%).

sum of conduction, switching, C_{oss}, dead time, inductor copper (from the manufacturer's REDEXPERT tool), capacitor ESR, gate drive, shunts, and the 3.3 V and 5 V rails predicts 1.98 W at point 2, or 98.5% (Fig. 9). Inductor copper is the largest term. GaN conduction and the capacitors are about 24% each. PCB copper is absent from the model and accounts for part of the gap to the bench.

B. Tracker tuning

The P&O parameters, update rate, step size, and power deadband, set the tracking speed, the steady-state oscillation, and the sensitivity to measurement noise. They were tuned in a MATLAB Simulink model of the boost converter, the 35-cell string, and the SG01 battery before the hardware existed. The high-side transistor is replaced by a diode, which simplifies the model and does not affect the algorithm comparison. The solar panel side measurements carry noise, as on the real board, therefore an averaged is executed over the last ten samples at 1 kHz. The algorithm runs at 100 Hz and includes the voltage sweep and the P&O loop.

The first question is how far the MPP moves. A duty-cycle sweep at several irradiances and temperatures shows that irradiance, from 1000 $\mathrm{W/m^2}$ to 200 $\mathrm{W/m^2}$, shifts the duty cycle by about 3% (at MPP), while temperature, from 25 °C to 85 °C, shifts it by about 9%. Temperature is therefore the disturbance that sets the required travel, even though irradiance changes faster. Temperature steps do not decrease 60 °C in such shot time.

At 100 Hz, fifty updates fit in the 0.5 s budget, so a 0.18% step would cover the 9% temperature travel, but a step that small is lost in the measurement noise. Compared on slow and fast changes, a 0.1% step is slow and can be confused with noise, a 2% step arrives quickly and oscillates around the MPP, and the useful range is 0.5–1%. The final values were set on the bench, because sensor error changes the step that the simulation suggests.

VI. Experimental Results

Three boards were fabricated by Eurocircuits and soldered by hand (Fig. 10). The bill of materials is about 90 € and the board 3–35 €, so a finished charger is about 125 €, against 240 € for the commercial unit. Most tests used version 1.1. The CISPR 25 scan and the outdoor charge used version 1.2,



Fig. 10. Front side of the three prototypes versions.

TABLE III
CISPR 25 conducted voltage limits, peak/average in dBμV, for the five EMC classes.

Band (MHz)	Class 1	Class 2	Class 3	Class 4	Class 5
0.15–0.3	113/100	103/90	93/80	83/70	73/60
0.53–2	95/82	87/74	79/66	71/58	63/50
5.9–6.2	77/64	71/58	65/52	59/46	53/40
30–54	77/64	71/58	65/52	59/46	53/40
68–108	61/48	55/42	49/36	43/30	37/24

which moved the sense filters next to the ADC and added the ceramics required by the noise tests.

A. Measurement error

From 0 to 51 V, the mean absolute voltage error was 60 mV on the battery divider and 112 mV on the panel divider. The error is nearly linear, and subtracting it in firmware brings both means to about 29 mV. The mean absolute current error was 100 mA on the battery shunt and 8.5 mA on the solar panel shunt, the same correction on the battery channel brings that mean to 5.3 mA. All measured imprecision do not have a meaningful affect on the algorithm.

B. Conducted emissions

The charger shares the vessel with a CAN bus and with radio equipment. Conducted noise on the supply lines is the path that reaches both, so the prototype was compared with CISPR 25:2021, the standard for radio disturbance from electronic components used on vehicles. The standard defines five limit classes for conducted voltage on the power lines. Class 5 is the most strict. Table III lists the peak and average limits of all five classes in the bands used for this comparison, from the long-wave range through the FM broadcast band.

The scan was made after the output decoupling had been corrected. The first board had a single 1 μF ceramic at the output, against ten 1 μF parts on the GaN evaluation board, and input-current spikes of about 1.8 A were tripping the overcurrent protection. Nine additional 1 μF capacitors cut that excursion to 640 mA, and six 100 nF capacitors brought it to about 290 mA. The spectral measurement was performed on that decoupled hardware.

The supply was set to 22 V, the load was 10 Ω, and the duty cycle was fixed at 52.5% (Fig. 11). The measurement

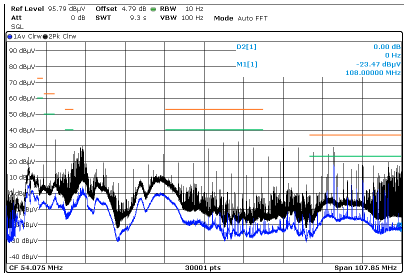


Fig. 11. Conducted spectrum with a 10 Ω load, from 22 V at $D = 52.5\%$. The trace stays below the CISPR 25 Class 5 limits in the bands of Table III.

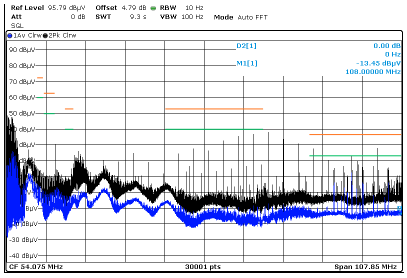


Fig. 12. Conducted spectrum with no external load and the output held at 50.4 V. The level is higher than in Fig. 11 and remains below the same Class 5 limits.

TABLE IV
MEASURED EFFICIENCY. THE LAST COLUMN SUBTRACTS THE 0.51 W CONTROL RAIL AND THE NO-LOAD SWITCHING OVERHEAD.

Case	P_{in} (W)	P_{out} (W)	η (%)	η power stage (%)
1	127.4	123.4	96.8	97.6
2	129.1	124.7	96.6	97.4
3	127.5	123.2	96.6	97.4
4	73.1	70.8	96.8	98.2
OC	0.5	0	0	0

was then repeated with the load removed and the output held at 50.4 V (Fig. 12). In both conditions the recorded level stays below the Class 5 peak and average limits of Table III in every listed band. The no-load spectrum sits higher. With little load current the switching waveform changes, and a larger share of the measured energy falls at the high-frequency end of the scan.

C. Efficiency and temperature

The efficiency was measured with a fix duty cycle and resistive load to match four charging points of the CC-CV curve present in simulation. The reading (Table IV) includes every loss on the board, sensing and control included. The end-to-end efficiency is 96.6-96.8%, which is above the 90% target. Considering a power consumption of 0.5 W with the DC-DC turned off, the efficiency of the boost converter is 97.4-98.2%. The remaining gap between simulation and experimental result is likely due to PCB copper, which simulations omit.

A second efficiency test was run with the MPPT algorithm active and connected to a battery and a photovoltaic emulator (SM 1500-CP-30). The emulator irradiance setting was slowly increased via TCP connection. The result (Fig. 13) show an

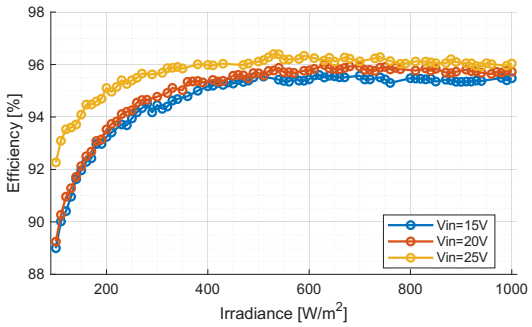


Fig. 13. Converter efficiency versus panel current for three string lengths at $V_{\text{bat}} = 43.4$ V.

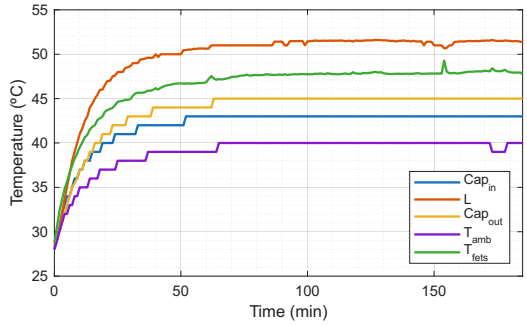


Fig. 14. Component temperatures during a 3 h run at charging point 3 inside a closed box.

increase in efficiency with irradiance/current, since the impact of the auxiliary circuit consumption is reduced. After around 3 A (400 W/m^2), the slopes slightly down as switching loss grows. Longer strings are more efficient at every irradiance point since the V_{in} increases, duty cycle decreases and therefore the efficiency increases.

The temperature of the components was analyzed during 3 h in a close box, at MPP in case 3 of the CC-CV curve. The results (Figure 14) showed an increase of temperature in the first 60 min and then stalled. All temperatures are within the safety limits, reaching only a peak of 51.6 °C. It is concluded that no force cooling is required.

D. Tracking and charging

Laboratory test using a Delta SM 1500-CP-30 in photovoltaic mode, configured as a 35-cell string with irradiance and temperature set over TCP, charging the 12s7p pack. The operator sets the environment, so each profile can be repeated. After bench tuning, the P&O loop samples at 1 kHz, updates at 50 Hz, steps 0.7% of duty cycle, and uses a 700 mW power deadband. The update is slower than the 100 Hz of the Simulink study so that each decision averages more samples, the step lies in the 0.5–1% window that study identified, however a power deadband was still needed to ignore the excessive noise identified in version 1.1. If the readings are within the deadband, the algorithm ignores it. In version 1.2 the senors noise was solved and the power deadband was removed.

The state machine was checked first. After some troubleshooting and tuning, the state machine behave as it is

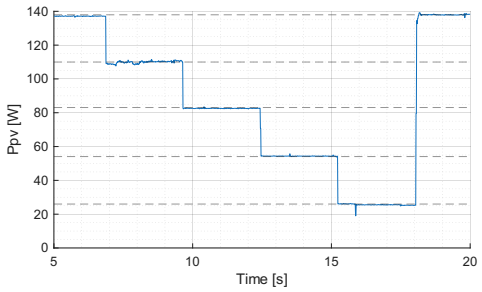


Fig. 15. Closed-loop power during the irradiance staircase. The tracker stays on the moving MPP and recovers the final step from 200 to 1000 $\mathrm{W/m^2}$ in a fraction of a second.

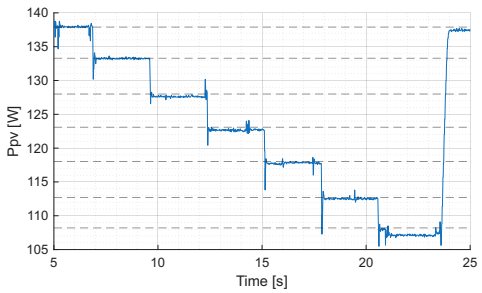


Fig. 16. Closed-loop power during the temperature staircase. The slowest recovery is the final step from 85 °C to 25 °C.

supposed. Only the sweep occasionally tripped overcurrent protection when used with the photovoltaic emulator. However, it works as expected with a solar panel.

Tracking speed was tested with two scripted profiles (Figs. 15 and 16). Irradiance fell from 1000 to 200 $\mathrm{W/m^2}$ in 200 $\mathrm{W/m^2}$ steps every 2 s, then jumped back in one step. Temperature rose from 25 to 85 °C in 10 °C steps every 2 s, then dropped back in one step. Gradual steps represent ordinary operation, the jumps are the worst case. Time to re-enter $\pm 5\%$ of the MPP power was below 100 ms on every gradual step and on the irradiance jump. The 85 to 25 °C jump, the slowest case in simulation, took about 200 ms. Average tracking accuracy was about 99%, with the largest offset at 85 °C, most likely from measurement noise. Both figures meet the 1 s limit and the 0.5 s typical target with margin.

Two full charges close the campaign, both into a 12s1p, 5.14 Ah pack chosen to keep the charge time short. From the emulator (Fig. 17), the charge followed CC–CV: P&O held the string at the MPP until the battery reached its voltage limit, the current then decayed at constant voltage, and the machine entered the finished state. The second charge was outdoors, on shore, from two panels in series (42 cells) on a cloudy day (Fig. 18). Panel current followed the changing irradiance, while panel voltage, set by a nearly constant temperature, moved only slightly to stay at the MPP. An interruption occurred due to accidentally disconnection of the solar panels. This run used revision 1.2, on which the lower noise made the 700 mW deadband too large for slow outdoor changes. The deadband was removed for the test. The charger has not yet been operated on SG01.

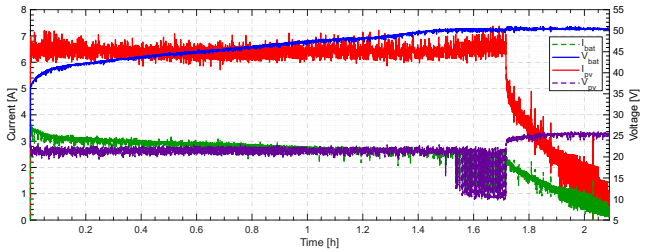


Fig. 17. Laboratory charge of the 12s1p pack from the photovoltaic emulator. Constant current is the panel MPP; constant voltage follows once the cell limit is reached.

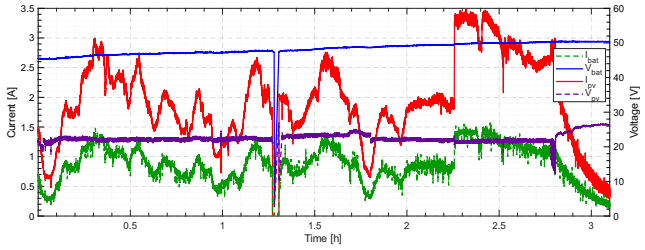


Fig. 18. Outdoor on-shore charge from two series panels (42 cells) on a cloudy day. Both the CC and CV intervals are visible.

VII. Conclusion

A fast-response MPPT charger was designed, built, and tested for the low-voltage auxiliary battery of SG01. The vessel's motion was used to establish the tracking requirements: following changes in irradiance or temperature, the operating point should return to the MPP within 1 s, with a target of 0.5 s under typical conditions. These requirements guided the development of a 100 kHz synchronous GaN boost converter, dimensioned using the electrical relationships governing the inductor and capacitors and subsequently refined through LTspice simulations. The power stage was combined with a P&O algorithm, initially tuned in Simulink and subsequently refined through laboratory testing.

The charger achieves an end-to-end efficiency of 96.6-96.8%, exceeding the target of 90%. Excluding the power consumption of the auxiliary electronics, the boost converter efficiency is approximately 97.4-98.2%. The tracking algorithm returns to within $\pm 5\%$ of the MPP power in less than 100 ms following typical irradiance and temperature changes, and within 200 ms following the largest temperature step tested.

Conducted emissions remain below the CISPR 25 Class 5 limits. Thermal testing in a closed enclosure showed that the hottest component reached a steady-state temperature of 51.6 °C without forced cooling.

Complete CC-CV charging cycles were demonstrated both in the laboratory and outdoors on shore, demonstrating the functionality of the prototype. The assembled charger costs approximately 125 €, compared with approximately 240 € for the commercial MPPT charger currently in use. In addition, the unit supports configuration through a graphical user interface over serial communication and communication with the vessel's systems via CAN.

Although the prototype is not yet a finished product, its functionality has been demonstrated, and it is suitable for installation on SG01, pending onboard validation. The main remaining step is testing aboard the vessel, where environmental variations may be more severe and less predictable than those encountered in laboratory and onshore tests. Following this validation, alternative MPPT algorithms should be implemented, tuned, and compared under equivalent operating conditions. Adding a dedicated irradiance sensor, alongside the temperature sensing already available on the board, would improve the estimation of conversion losses and provide additional information to support the tracking algorithms. Further improvements could include a microcontroller with a native 16-bit ADC to increase measurement resolution and a higher clock frequency to improve PWM duty-cycle resolution. Finally, interleaving the four chargers that share the battery could reduce both component costs and output-voltage ripple.

REFERENCES

[1] Our World in Data, "Electricity mix," 2025, accessed: 2025-11-24. [Online]. Available: https://ourworldindata.org/electricity-mix

[2] U. Shahira, S. Z. Islam, R. b. Omar, M. L. Othman, S. Z. Said, and J. Uddin, "Electrical design of solar-powered recreational boat in malaysia," in *2022 IEEE International Conference on Power and Energy (PECon)*, 2022, pp. 138–143.

[3] International Council on Clean Transportation (ICCT), "Transport could burn up the eu's entire carbon budget," n.d., accessed: 2025-11-24. [Online]. Available: https://theicct.org/transport-could-burn-up-the-eus-entire-carbon-budget/

[4] H. Häberlin, *Photovoltaics: system design and practice*. John Wiley & Sons, 2012.

[5] *New Frontiers in Operational Oceanography*. CreateSpace Independent Publishing Platform, 2018, gOADE OceanView.

[6] T. Esram and P. L. Chapman, "Comparison of photovoltaic array maximum power point tracking techniques," *IEEE Transactions on Energy Conversion*, vol. 22, no. 2, pp. 439–449, 2007.

[7] B. Subudhi and R. Pradhan, "A comparative study on maximum power point tracking techniques for photovoltaic power systems," *IEEE Transactions on Sustainable Energy*, vol. 4, no. 1, pp. 89–98, 2013.

[8] D. Hohm and M. Ropp, "Comparative study of maximum power point tracking algorithms using an experimental, programmable, maximum power point tracking test bed," in *Conference Record of the Twenty-Eighth IEEE Photovoltaic Specialists Conference - 2000 (Cat. No.00CH37036)*, 2000, pp. 1699–1702.

[9] B. M. Ali Chermitti, Omar Boukli-hacene, "Improvement of the ìperturb and observeî mppt algorithm in a photovoltaic system under rapidly changing climatic conditions," *International Journal of Computer Applications*, vol. 56, no. 12, pp. 11–16, October 2012.

[10] J. K. Udavalakshmi and M. S. Sheik, "Comparative study of perturb & observe and look-up table maximum power point tracking techniques using matlabisimulink," in *2018 International Conference on Current Trends towards Converging Technologies (ICCTCT)*, 2018, pp. 1–5.

[11] O. Wasynezuk, "Dynamic behavior of a class of photovoltaic power systems," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-102, no. 9, pp. 3031–3037, 1983.

[12] E. Roman, R. Alonso, P. Ibanez, S. Elorduizapatarietxe, and D. Goitia, "Intelligent pv module for grid-connected pv systems," *IEEE Transactions on Industrial Electronics*, vol. 53, no. 4, pp. 1066–1073, 2006.

[13] B. Bendib, H. Belmili, and F. Krim, "A survey of the most used mppt methods: Conventional and advanced algorithms applied for photovoltaic systems," *Renewable and Sustainable Energy Reviews*, vol. 45, pp. 637–648, 2015.

[14] W. Shen, T. T. Vo, and A. Kapoor, "Charging algorithms of lithium-ion batteries: An overview," in *2012 7th IEEE Conference on Industrial Electronics and Applications (ICIEA)*, 2012, pp. 1567–1572.

[15] A. Al-Haj Hussein and I. Batarseh, "A review of charging algorithms for nickel and lithium battery chargers," *IEEE Transactions on Vehicular Technology*, vol. 60, no. 3, pp. 830–838, 2011.

[16] J. Fattal and P. B. Dib Nabil Karami, "Review on different charging techniques of a lithium polymer battery," in *2015 Third International Conference on Technological Advances in Electrical, Electronics and Computer Engineering (TAEECE)*, 2015, pp. 33–38.

[17] S. Lee and H. Do, "High-efficiency soft-switching step-up dc-dc converter derived from a synchronous boost converter," *IET Power Electronics*, vol. 12, 06 2019.

[18] S. Talat. (2024, Oct.) Igbt, mosfet and gan: An overview of efficiency, power and system cost for inverter design. Taiwan Semiconductor. Accessed: 2026-07-19. [Online]. Available: https://www.taiwansemi.com/en/blog-igbt-mosfet-gan/

[19] V. Lo Brano and G. Ciulla, "An efficient analytical approach for obtaining a five parameters model of photovoltaic modules using only reference data," *Applied Energy*, vol. 111, pp. 894–903, 2013. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261913005539